THREE ORDINAL-INDEXED COMPARISONS OF PRODUCTS WITH ITERATED Z HOLD AT EVERY COUNTABLE ORDINAL

DAVID GONZALEZ, DINO ROSSEGGER

Editorial note. The counted statement is the author environment at source0.tex lines 1147-1158 and its proof environment opens at line 1160, the line immediately after; the pairing is by adjacency and the retained proof range 1160-1266 is the whole environment.

Editorial note. The retained preamble is the author style file dino.sty (package lines, the theorem declaration support, the notation and tuple macros and the theorem environments) followed by the ordinal and rank macros of the manuscript itself, reproduced verbatim; this is the same recipe as the previously published card of this article, so no package outside it is requested and extra_packages stays empty.

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Editorial note. Three item labels are defined inside the retained statement, one for each family A_alpha, B_alpha and C_alpha, and the retained proof refers to the families by those names; they need no external entry. No context span and no external reference are needed: the retained statement and proof use no label or citation from outside their own ranges.

Editorial scope. One original result of the pinned author manuscript, counted once: the three ordinal-indexed comparison families for products with iterated Z at the ranks two alpha, two alpha plus one and two alpha plus two (author label lem:timesZ).

Allowed standard prerequisites (editorial).

- Iterated products with Z. Z^alpha is the alpha-fold lexicographic product of the integers with itself, and Z^alpha times L is the product of that order with L.. They are the orders the three families of statements compare.
- Back-and-forth relations and their rank shift. The rank two alpha, two alpha plus one and two alpha plus two relations are decided by intervals of the products, whose form fixes how many moves the comparison costs.. They are the relations the three statements assert at those ranks.
- Initial and final segments of the products. An initial segment of Z^alpha times L above a chosen point is again a product with an infinite order, and the same holds for final segments.. They are the orders appearing in the interval cases the proof checks.

Lemma 0.1. *For the sake of organization, consider the following ordinal indexed propositions.*

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(A_α) *For any* $\mathcal{K}$ *and* $\mathcal{L}$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha} \mathbb{Z}^\alpha \cdot \mathcal{L}$.
(B_α) *For any* $\mathcal{K}$ *and* $\mathcal{L}$ *with* $|\mathcal{K}| \geq |L|$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}$.
(C_α) *For any* $\mathcal{K}$ *and any* $\mathcal{L}$ *without a last element,* $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \leq_{2\alpha+2} \mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$.

For any countable ordinal α, A_α, B_α *and* C_α *are true.*

Proof. The proof is by induction with the statements A_0 and B_0 trivially true. We start by showing that B_α implies $A_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{L}$. We will find a tuple $b_0 \dots b_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{K}$ such that $(\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}, \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}, \bar{b})$. Pick a point b_0 arbitrarily. Given b_j, we pick b_{j+1} as follows. If a_j and a_{j+1} are in the same copy of $\mathbb{Z}^{\alpha+1}$, we can pick b_{j+1} such that $(a_j, a_{j+1}) \cong (b_j, b_{j+1})$. If not, we pick b_{j+1} arbitrarily in the copy of $\mathbb{Z}^\alpha$ that is the successor of the successor of the copy of b_j. In other words, we pick b_{j+1} with $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha$.

To show that our choice of $\bar{b}$ is correct, we have to confirm the following relations between non-isomorphic intervals.

(1) $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ is an infinite order,
(2) $(-\infty, b_0) \cong \mathbb{Z}^\alpha \cdot \mathcal{K}_1 + \zeta_\alpha \geq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some initial segments of the $\mathbb{Z}^\alpha$, and therefore infinite, or
(3) $(b_n, \infty) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{K}_1$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some final segments of the $\mathbb{Z}^\alpha$, and therefore infinite.

It follows immediately from B_α that all of these back-and-forth relations hold.

We now show that B_α implies C_α. So assume that $\mathcal{K}$ and $\mathcal{L}$ have no last element and that B_α holds. Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$. Without loss of generality, a_0 is in the first copy of $\mathbb{Z}^\alpha$. We will find a tuple $b_0 \dots b_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K})$ such that $(\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L}), \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}), \bar{b})$. For the a_j in the initial $\mathbb{Z}^\alpha \cdot \omega$, the b_j are picked isomorphically. We now define the rest of the b_j by induction so that all of them are in the initial $\mathbb{Z}^\alpha \cdot \omega$. If a_j is in the same $\mathbb{Z}^\alpha$ block as a_{j+1} or a_{j+1} is in the succesor of a_j's block, define b_{j+1} so that $(b_j, b_{j+1}) \cong (a_j, a_{j+1})$. Otherwise, define b_{j+1} as being some element in the successor of the successor of the $\mathbb{Z}^\alpha$ block of b_j. Not including isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$ where $\mathcal{L}_1$ is infinite,
(2) $(b_n, \infty) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1$, where $\mathcal{L}_1$ is some final segment of $\omega + \mathcal{L}$, and thus infinite.

All of these relations follow from B_α, as desired.

To complete the proof of the successor step of the induction, we show that C_α and $A_{\alpha+1}$ imply $B_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$. These points belong to at most n distinct copies of $\mathbb{Z}^{\alpha+1}$. Because $|\mathcal{K}| \geq |\mathcal{L}|$, we can match each chosen copy of $\mathbb{Z}^{\alpha+1}$ to one in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$. Furthermore, if there is a copy of $\mathbb{Z}^{\alpha+1}$ between two of the chosen copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$, we can assure that the corresponding copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$ are also not successors. Also, if a_n is not in the last copy of $\mathbb{Z}^{\alpha+1}$, we can assure that neither is b_n and, similarly for a_0 and b_0 in the first copy. In particular, not including the isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(2) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(3) $(-\infty, b_0) \cong \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(4) $(-\infty, b_0) \cong \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some order $\mathcal{L}_1$,
(5) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by $A_{\alpha+1}$. By symmetry it is sufficient to show case (5). However, this is the same as showing that

$$\zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \omega \leq_{2\alpha+2} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathbb{Z} \cdot \mathcal{K}_1).$$

This follows directly from C_α.

At last we consider the remaining limit levels. We already have seen that B_λ implies C_λ. The statement A_λ follows immediately from A_γ for $\gamma < \lambda$. A bit more nuanced is the issue of B_λ. It follows from A_λ and C_γ for $\gamma < \lambda$. Analogous to the proof in the successor case, we analyze the possibilities:

(1) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(2) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(3) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(4) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda$ for some order $\mathcal{K}_1$,
(5) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by A_λ. By symmetry it is clear that it is enough to show case 5. However, this is the same as showing that for all $\gamma < \lambda$,

$$\zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\omega + \mathbb{Z} \cdot \zeta_\lambda^*) \leq_\gamma \zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\mathbb{Z} \cdot \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1),$$

which follows from the C_γ for $\gamma < \lambda$.

Therefore, for any countable ordinal α, A_α, B_α and C_α all hold. □